

Alan Ning

Software Engineer / Problem Solver

📍 Herndon, United States

✉ alan@askldjd.com

📞 (808) 232-7868

🔗 askldjd.com

👤 askldjd.com (Personal Blog)

👤 github.com/askldjd (Github Profile)

Work **Software Architect**

Manifold.ai

Oct 2021 – Present

I architect ML pipelines and generate meaningful outcome.

Staff Reliability Engineer

Tenable

Oct 2018 – Oct 2021

I create scalability solutions for Tenable.io.

- Designed and implemented a custom Elasticsearch sharding layer to break out a petabytes-scale monolith database into an infinitely scalable number of databases.
- Implemented a caching layer for the Tenable.io edge using Cloudflare Worker. This layer handles over 5000 requests per sec, and saves petabytes of bandwidth per year.

Site Reliability Engineer

Uptime2020 Consultant

Jun 2020 – Sep 2020

Supported various election-related SaaS companies to ensure their product security and reliability during the 2020 US election cycle.

Site Reliability Engineer

United States Digital Service - Department of Veterans Affairs

Sep 2016 – Aug 2018

I began as the SRE for Caseflow and left having architected the VA's migration to AWS.

- Guided the VA cloud migration effort, which includes Direct Connect configurations, network performance analysis, and VPC design that scales for thousands of tenants.
- Led the SRE team for the Veterans Appeals modernization effort (Caseflow).
- Designed and built a resilient cloud infrastructure for Caseflow that integrated with the VA's legacy systems.
- Migrated the 30 year old Oracle database that holds all Veterans Appeals to the cloud using AWS DMS.
- Automated my job away.

Technical Lead

United States Digital Service - Social Security Administration

Nov 2015 – Sep 2016

I began as a React/Node.js coach and left as the Technical Lead on the Disability Case Processing System (DCPS).

- Led and developed the Social Security Disability modernization project using microservice architecture with an HTML5 SPA frontend.
- Integrated the modern microservice architecture with legacy mainframe systems and mailing machines.
- Delivered a highly visible MVP under a tight deadline, which led to successful product shipment across many states.

Technical Lead Engineer

Boeing / Digital Receiver Technology

Jun 2005 – Oct 2015

I began as an embedded C++ developer and left as a full stack software developer.

- Wrote high performance C++ code with the focus of minimizing memory leaks, stack/heap corruptions, or race conditions.
- Developed multiple applications, both native and web-based, including technologies such as Java SOAP, C# WinForms, CassandraDB, Android, NVIDIA CUDA Framework and Node/React/AngularJS.
- Performed cybersecurity research that involved forwarding mobile IP packets using ARP poisoning.
- Became a technical lead for a wide variety of projects.

Education

Johns Hopkins University

Master in Computer Science

Sep 2005 – Jun 2007

Rensselaer Polytechnic Institute

Bachelor in Computer Science / Economics (Summa Cum Laude)

Sep 2001 – May 2005

Skills

Front-end Development

AngularJS

React

Vue.js

Server Side

Development

Node.js

Express.js

Java

C#

Python

Redis

Elasticsearch

Zookeeper

Ruby on Rails

Kafka

PostgreSQL

Cloud Development

Docker

Openshift

Kubernetes

AWS

Azure

Ansible

Terraform

Jenkins

Prometheus

Sentry

Redis

High Performance

Embedded Development

Modern C++

C

Mobile Development

Android